# Contents

Bandit problems
- Introduction
- Bernoulli bandits
- The unknown reward case
- Decision-theoretic bandit process
- Bernoulli-Beta conjugate pair

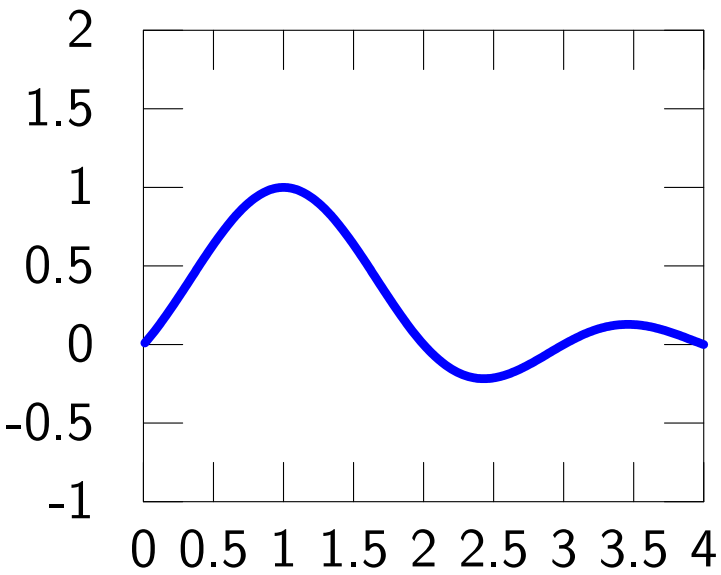
Markov decision processes and reinforcement learning
- Markov processes
- Markov decision processes

# Bandit problems

# Bandit problems

## Applications

- Efficient optimisation.



# Bandit problems

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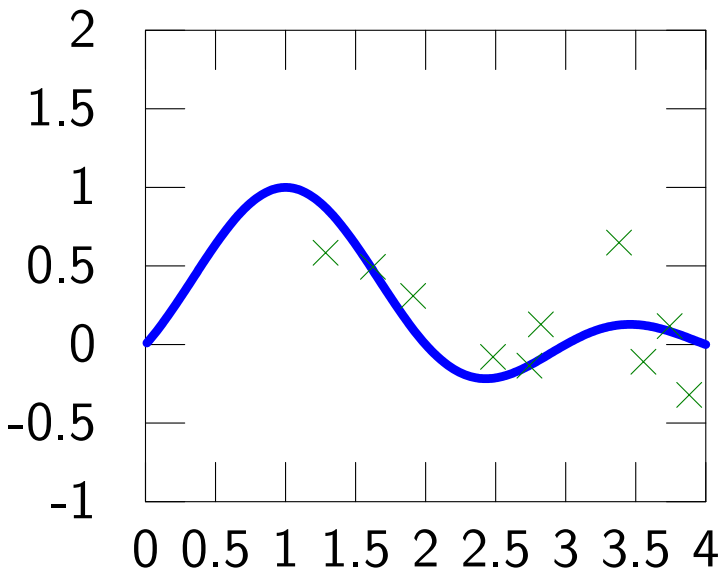

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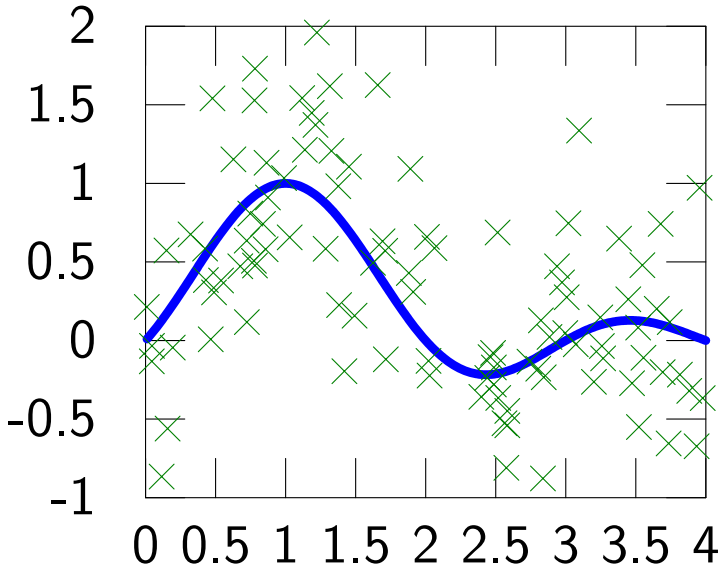

# Bandit problems

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- Efficient optimisation.
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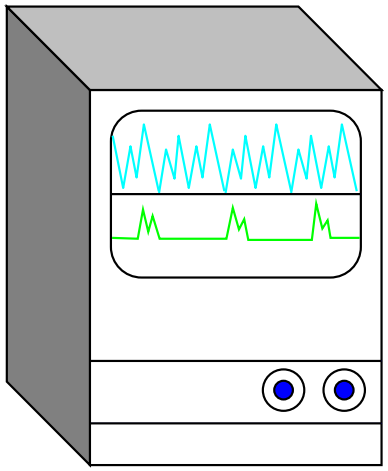

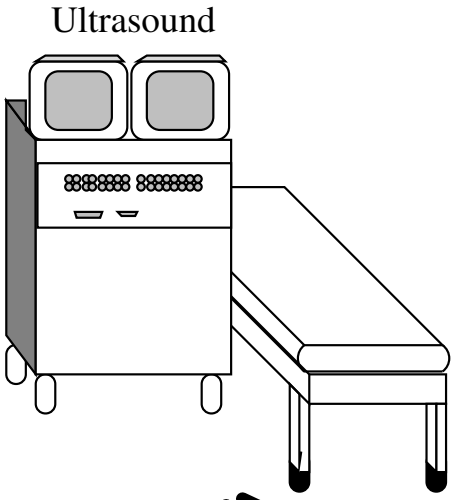


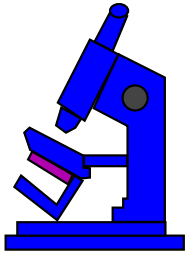

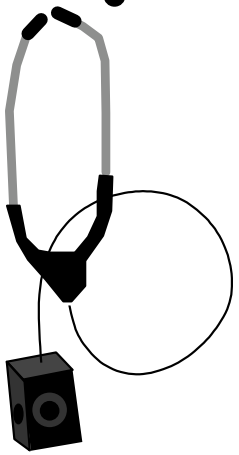

# Bandit problems

## Applications

- Efficient optimisation.
- Online advertising.
- Clinical trials.
- Robot scientist.

# The stochastic $n$-armed bandit problem

- A set of actions $\mathcal{A} = \{1, \dots, n\}$.
- The $i$-th action gives you a random reward

$$P_i(R) \triangleq \mathbb{P}(r_t \in R \mid a_t = i)$$

- The expected reward of the $i$-th arm is

$$\rho_i \triangleq \mathbb{E}(r_t \mid a_t = i)$$

## Utility

Given a reward sequence $r^T = (r_1, \dots, r_T)$, the utility is the total reward

$$U(r^T) \triangleq \sum_{t=1}^{T} r_t.$$

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## Utility

Given a reward sequence $r^T = (r_1, \dots, r_T)$, the utility is the total reward

$$U(r^T) \triangleq \sum_{t=1}^{T} r_t.$$

How should we select actions?

# Policy

## Definition 1 (Policies)

A policy $\pi$ is an algorithm for taking actions given the observed history.

$$\mathbb{P}^{\pi}(a_{t+1} \mid a_1, r_1, \ldots, a_t, r_t)$$

is the probability of the next action $a_{t+1}$.

# Bernoulli bandits

## Example 2 (Bernoulli bandits)

Consider $n$ Bernoulli distributions with parameters $\mu_i$ $(i = 1, \ldots, n)$ such that $r_t \mid a_t = i \sim \mathcal{Bern}(\mu_i)$. Then,

$$\mathbb{P}(r_t = 1 \mid a_t = i) = \mu_i \qquad \mathbb{P}(r_t = 0 \mid a_t = i) = 1 - \mu_i \tag{1.1}$$

Then the expected reward for the $i$-th bandit is $\rho_i \triangleq \mathbb{E}(r_t \mid a_t = i) =?$.

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Then the expected reward for the $i$-th bandit is $\rho_i \triangleq \mathbb{E}(r_t \mid a_t = i) = \mu_i$.

## Exercise 1 (The optimal policy under perfect knowledge)

*If we know $\mu_i$ for all $i$, what is the best policy?*

A *At every step, choose $i$ with the greatest $\mu_i$.*

B *At every step, choose $i$ with probability increasing with $\mu_i$.*

C *There is no right answer. It depends on the horizon $T$.*

D *Something else.*

E *I have no idea, it is too complicated.*

## Example 3 (The one-armed bandit problem)

Actions $\mathcal{A} = \{\text{stop}, \text{play}\}$.

$$\mathbb{P}(r_t = 1 \mid a_t = \text{play}) = \omega, \qquad \mathbb{P}(r_t = -1 \mid a_t = \text{play}) = 1 - \omega \tag{1.2}$$

If we stop, then $r_t = 0$, we get no more reward and the game ends. We must stop after $T$ plays.

If we know that $\omega = \omega^*$, and the horizon is $T$, what is the optimal strategy?

## Example 3 (The one-armed bandit problem)

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If we stop, then $r_t = 0$, we get no more reward and the game ends. We must stop after $T$ plays.

If $\omega$ is unknown but we believe $\omega \in \left\{0, \frac{1}{2}, 1\right\}$, and $\xi(\omega) = 1/3$, and $T = 1$, what is the subjectively ($\xi$)-optimal strategy?

A Play once.

B Stop immediately.

C Both strategies are optimal.

D Some other strategy is optimal (specify which).

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If we think $\omega \in \Omega = \left\{0, \frac{1}{2}, 1\right\}$, and $\xi(\omega) = 1/3$, and $T = 2$, how can we construct a subjectively optimal strategy?

## Example 3 (The one-armed bandit problem)

Actions $\mathcal{A} = \{\text{stop}, \text{play}\}$.

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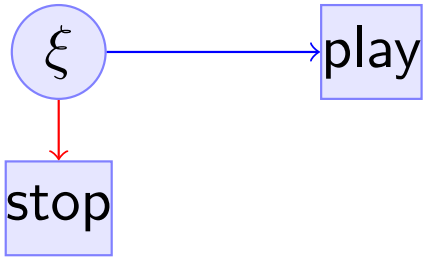
If we think $\omega \in \Omega = \left\{0, \frac{1}{2}, 1\right\}$, and $\xi(\omega) = 1/3$, and $T = 2$, how can we construct a subjectively optimal strategy? Hint: What is the set of possible strategies?

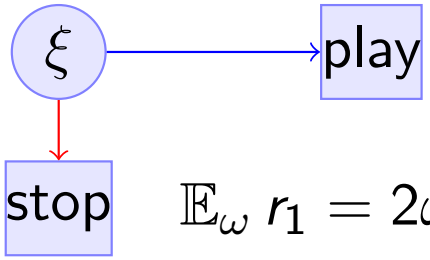



What is the $\xi$-expected reward $r_1$ at the first step? What about the actual expected reward?

$\xi_1$ $a_1$

Idea: construct a decision tree

$$\mathbb{E}_\xi\, r_1 = 0$$




$$\mathbb{E}_\omega\, r_1 = 2\omega - 1$$

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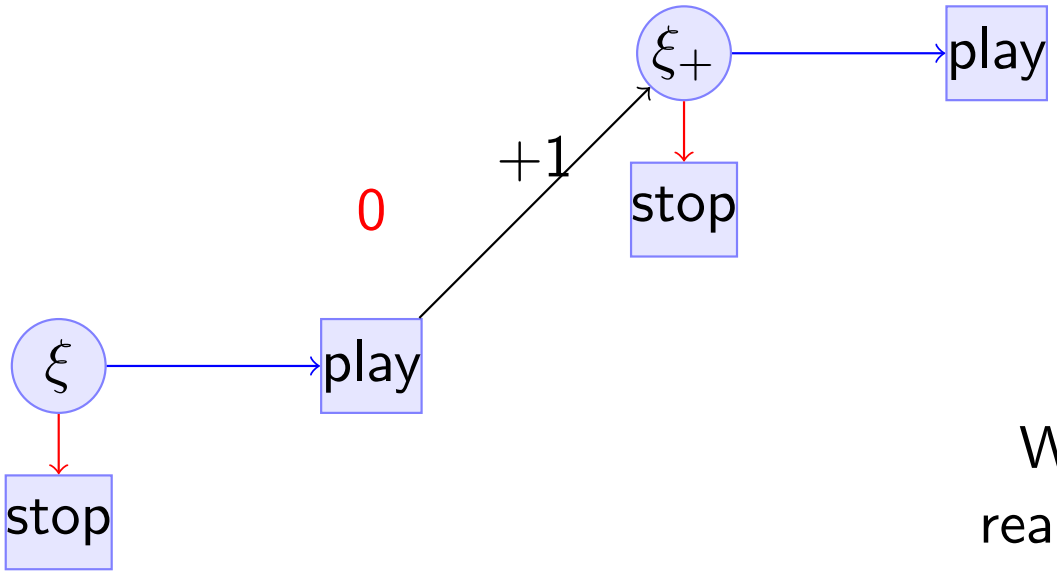


What is the $\xi$-probability of reaching belief $\xi_+$ at the next step?

$\xi_1$ $a_1$ $r_1$ $\xi_2$ $a_2$

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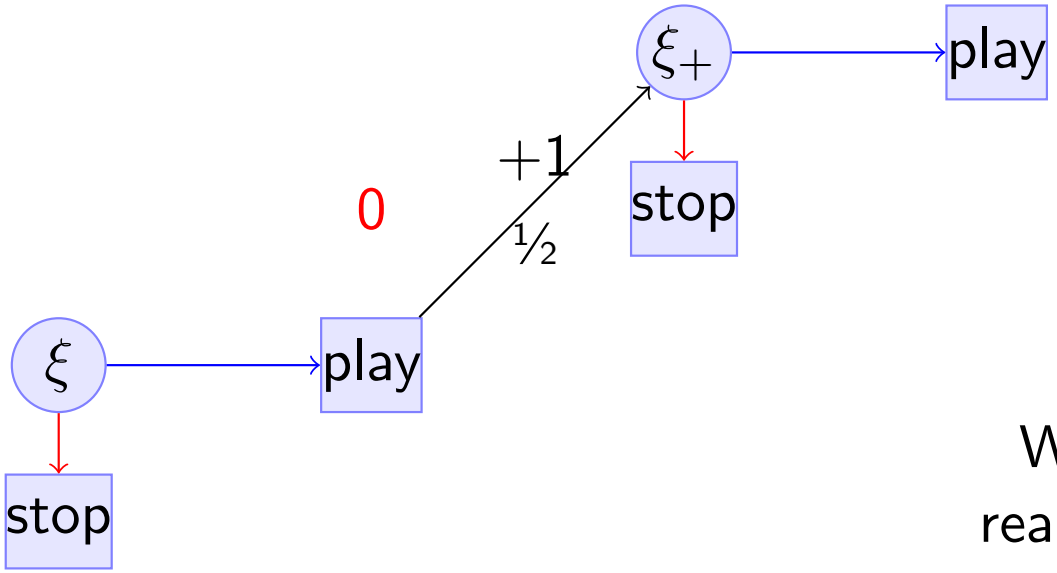


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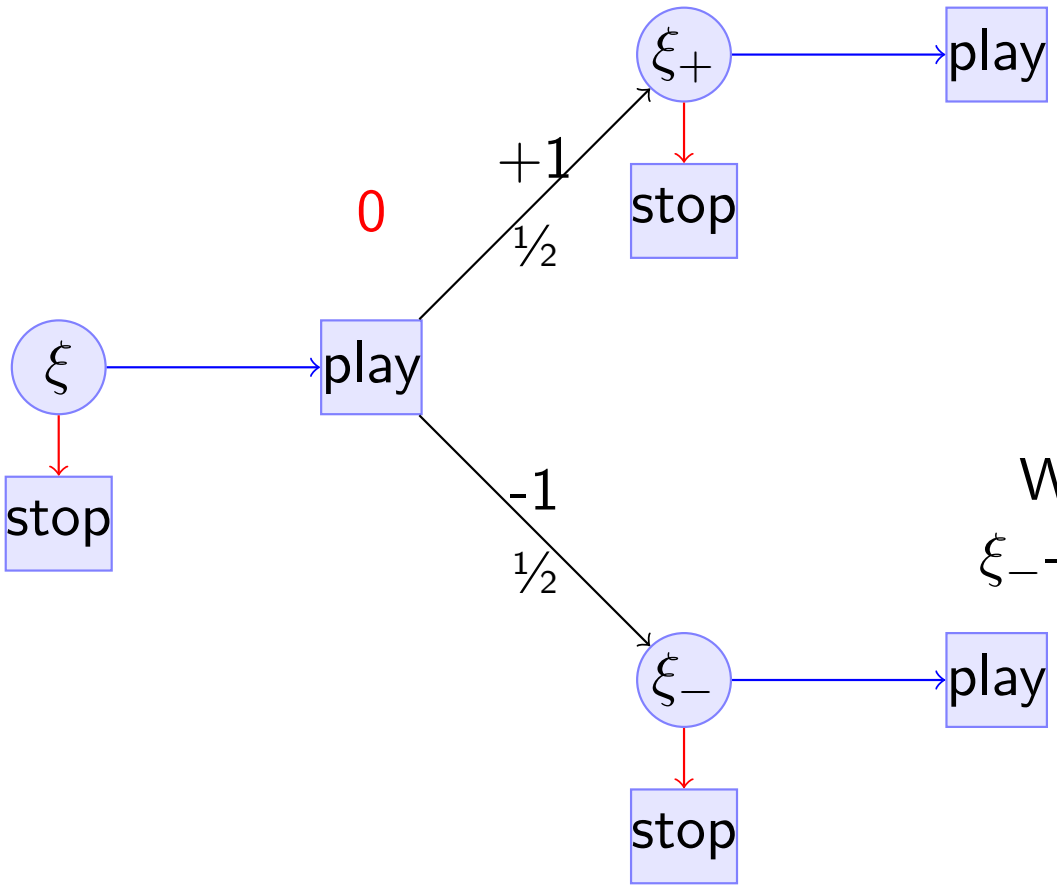
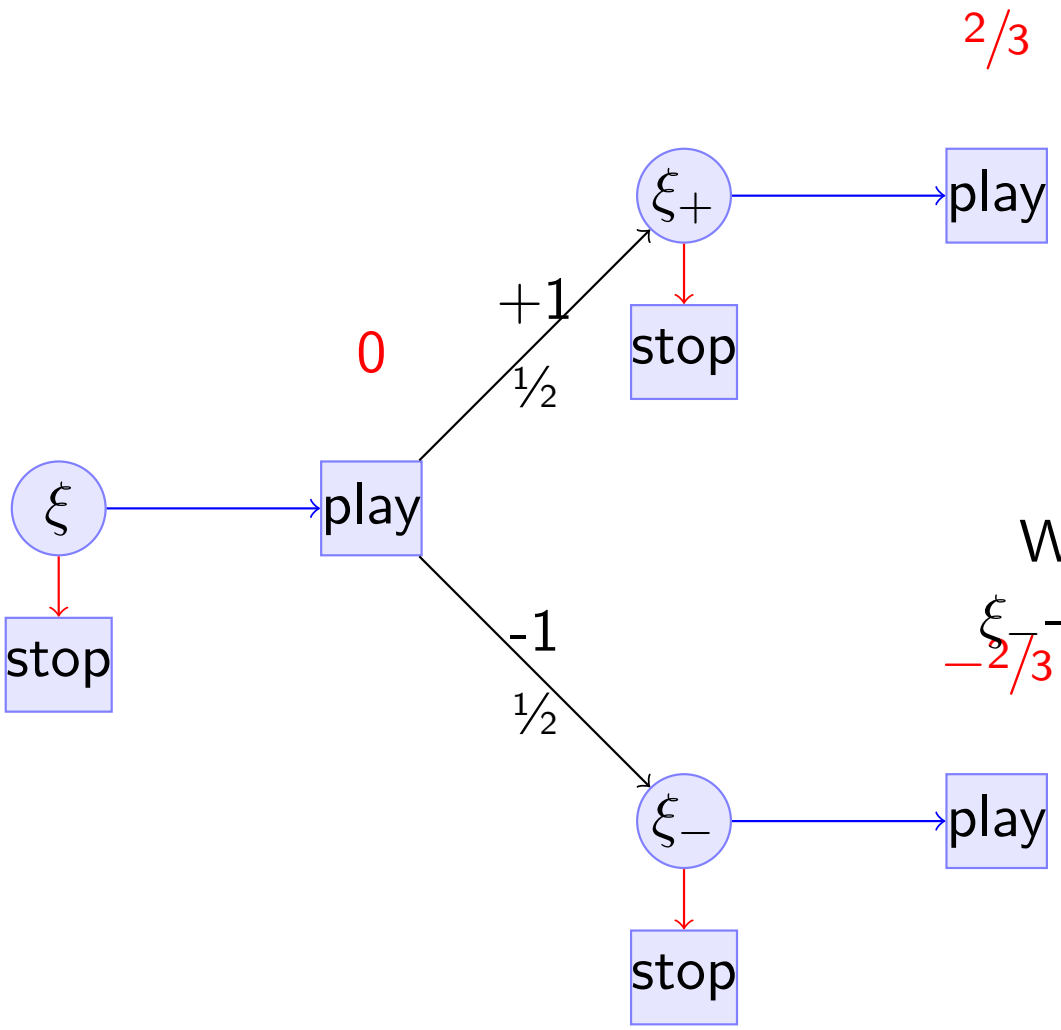
$$\mathbb{P}_\xi(r_1 = 1 \mid a_1 = \text{play}) = \sum_\omega \mathbb{P}_\omega(r_1 = 1 \mid a_1 = \text{play})\xi(\omega) = 1/2.$$




What are the $\xi_+$-expected and $\xi_-$-expected rewards?

$\xi_1$ $a_1$ $r_1$ $\xi_2$ $a_2$

Calculate rewards and utilities as belief changes

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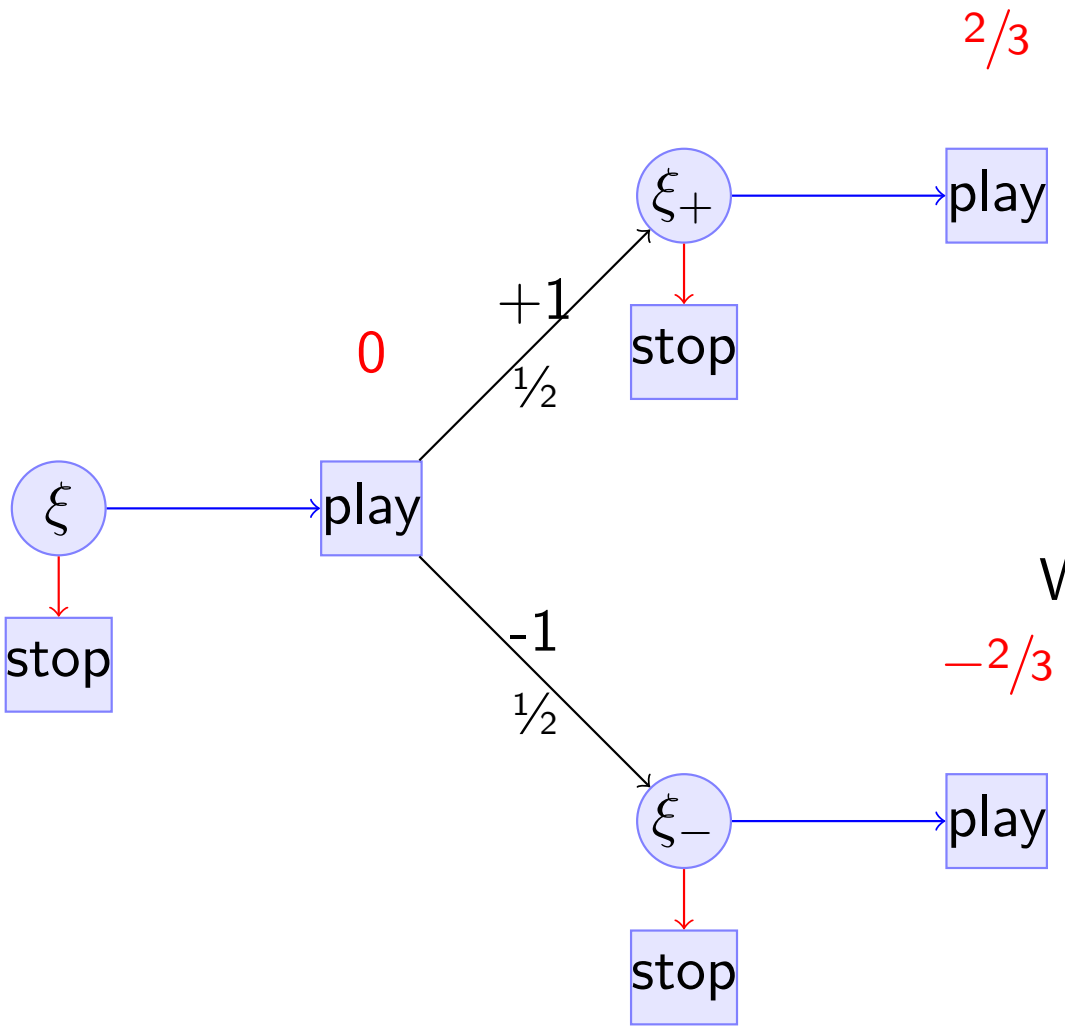
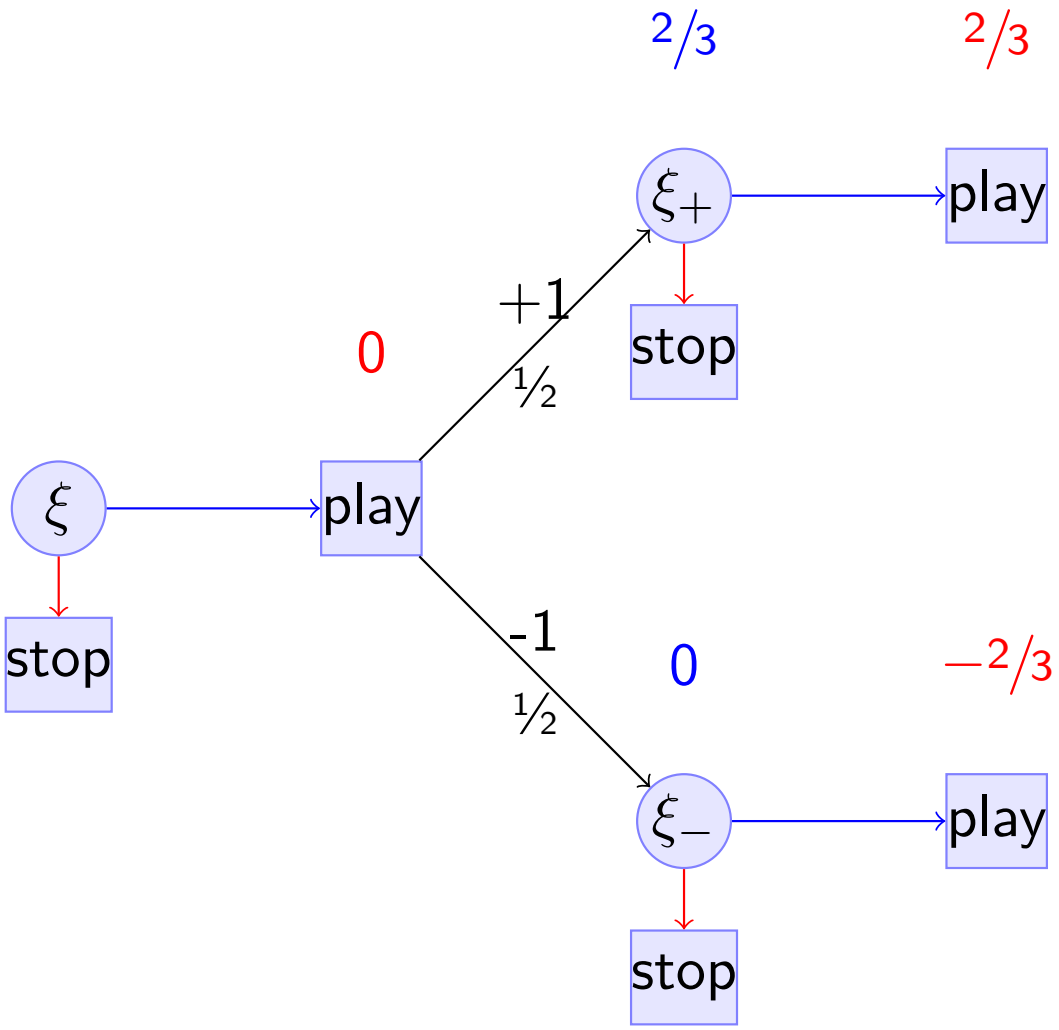



What is the $\xi$-optimal policy?

$\xi_1$ $a_1$ $r_1$ $\xi_2$ $a_2$

Calculate rewards and utilities as belief changes

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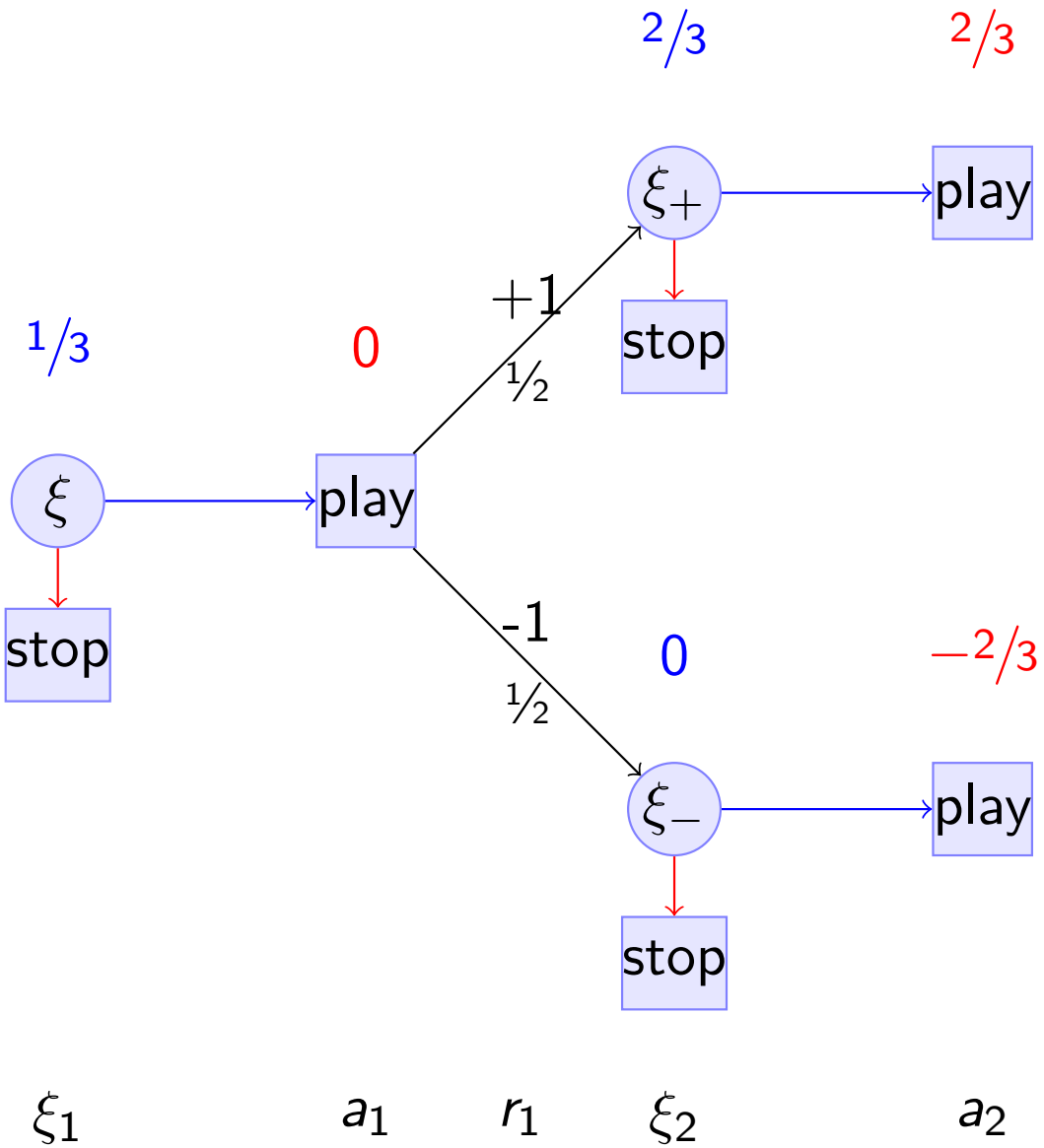


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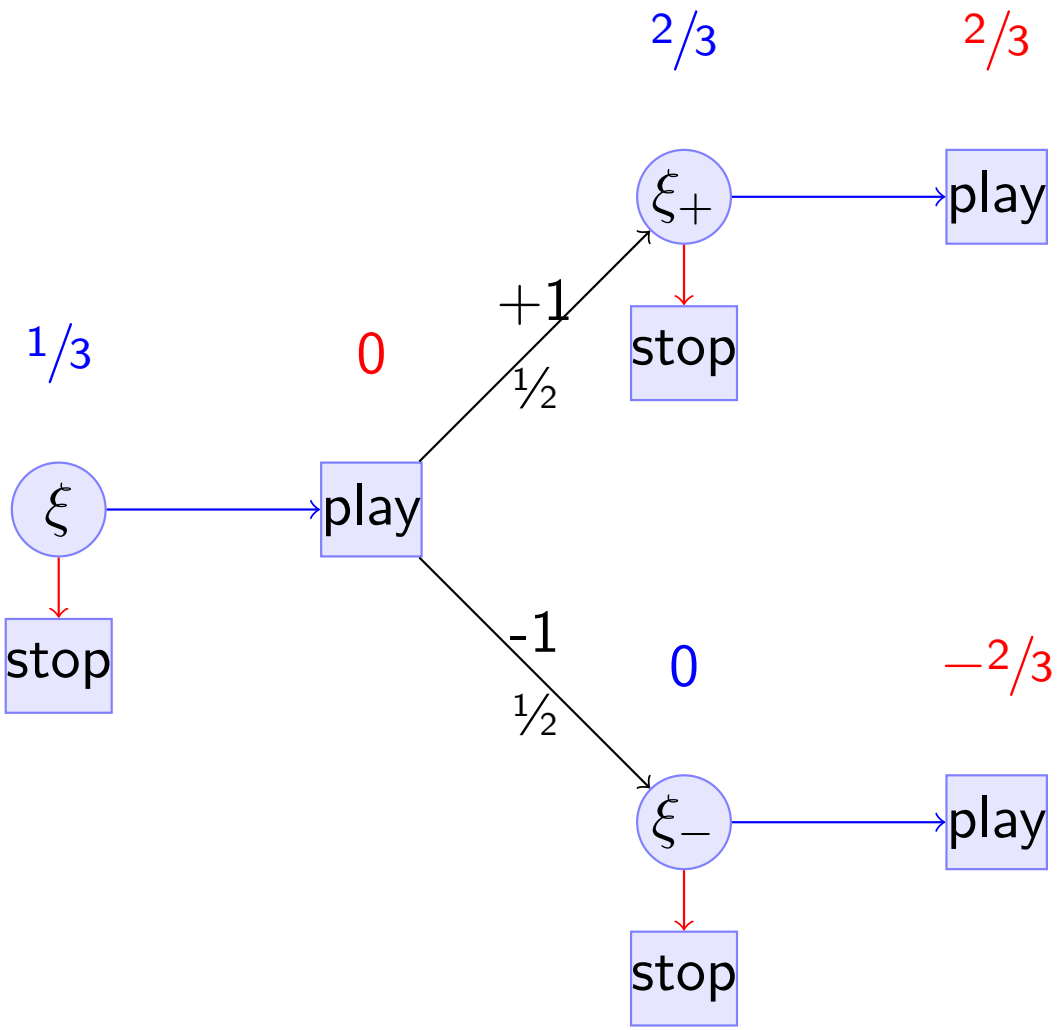


$\xi_1$ $a_1$ $r_1$ $\xi_2$ $a_2$

$$U^*(\xi_t) = \max_{a_t} \mathbb{E}(r_t \mid \xi_t, a_t) + \sum_{\xi_{t+1}} \mathbb{P}(\xi_{t+1} \mid \xi_t, a_t) U^*(\xi_{t+1}).$$

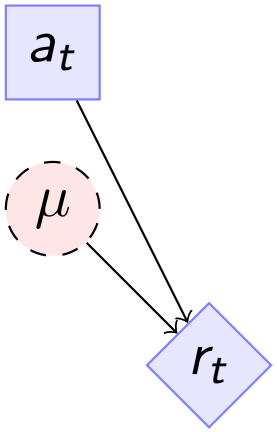


Figure: The basic bandit process. The decision maker selects $a_t$, while the parameter $\mu$ of the process is hidden. It then obtains reward $r_t$. The process repeats for $t = 1, \ldots, T$.

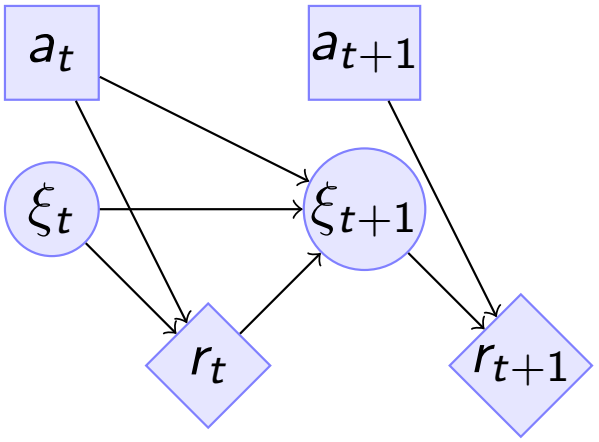


Figure: The decision-theoretic bandit process. While $\mu$ is not known, at each time step $t$ we maintain a belief $\xi_t$ on $\mu$. The reward distribution is then defined through our belief.

# Back to the general case

## Example 4 (Bernoulli bandits)

Consider $n$ Bernoulli distributions with parameters $\mu_i$ $(i = 1, \dots, n)$ such that $r_t \mid a_t = i \sim \mathcal{Bern}(\mu_i)$. Then,

$$\mathbb{P}(r_t = 1 \mid a_t = i) = \mu_i \qquad \mathbb{P}(r_t = 0 \mid a_t = i) = 1 - \mu_i \tag{1.3}$$

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Then the expected reward for the $i$-th bandit is $\rho_i \triangleq \mathbb{E}(r_t \mid a_t = i) = \mu_i$.

## The optimal policy when $\mu$ is unknown.

When we have a subjective belief $\xi$, the same idea can be used, in principle. There are two problems:

- It might not always be possible to compute beliefs.
- The decision-theoretic process has size $O(d^T)$, where $d$ can be $\infty$.

# Bernoulli distribution

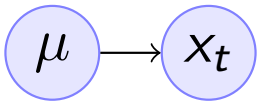


Figure: Bernoulli graphical model

## Definition 5 (Bernoulli distribution)

The Bernoulli distribution is discrete, with outcomes $\mathcal{S} = \{0, 1\}$, prameter $\mu \in [0, 1]$, and probability function:

$$P_\mu(u) = \begin{cases} \mu, & u = 1 \\ 1 - \mu, & u = 0 \end{cases} = \mu^u (1 - \mu)^{1-u}.$$

If $x_t$ is distributed according to a Bernoulli distribution with parameter $\mu$, we write $x_t \sim \mathcal{B}ern(\mu)$.

# Beta distribution

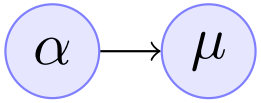


Figure: Beta graphical model

The Beta distribution has outcomes $\mu \in \mu = [0, 1]$ and parameters $\alpha_0, \alpha_1 > 0$, $\alpha = (\alpha_1, \alpha_0)$ with density:

$$p(\mu \mid \alpha) = \frac{\Gamma(\alpha_0 + \alpha_1)}{\Gamma(\alpha_0)\Gamma(\alpha_1)} \times \mu^{\alpha_1 - 1}(1 - \mu)^{\alpha_0 - 1}, \tag{1.4}$$

If $\mu \sim \mathcal{Beta}(\alpha_1, \alpha_0)$ then $\mathbb{E}\,\mu = \frac{\alpha_1}{\alpha_0 + \alpha_1}$.

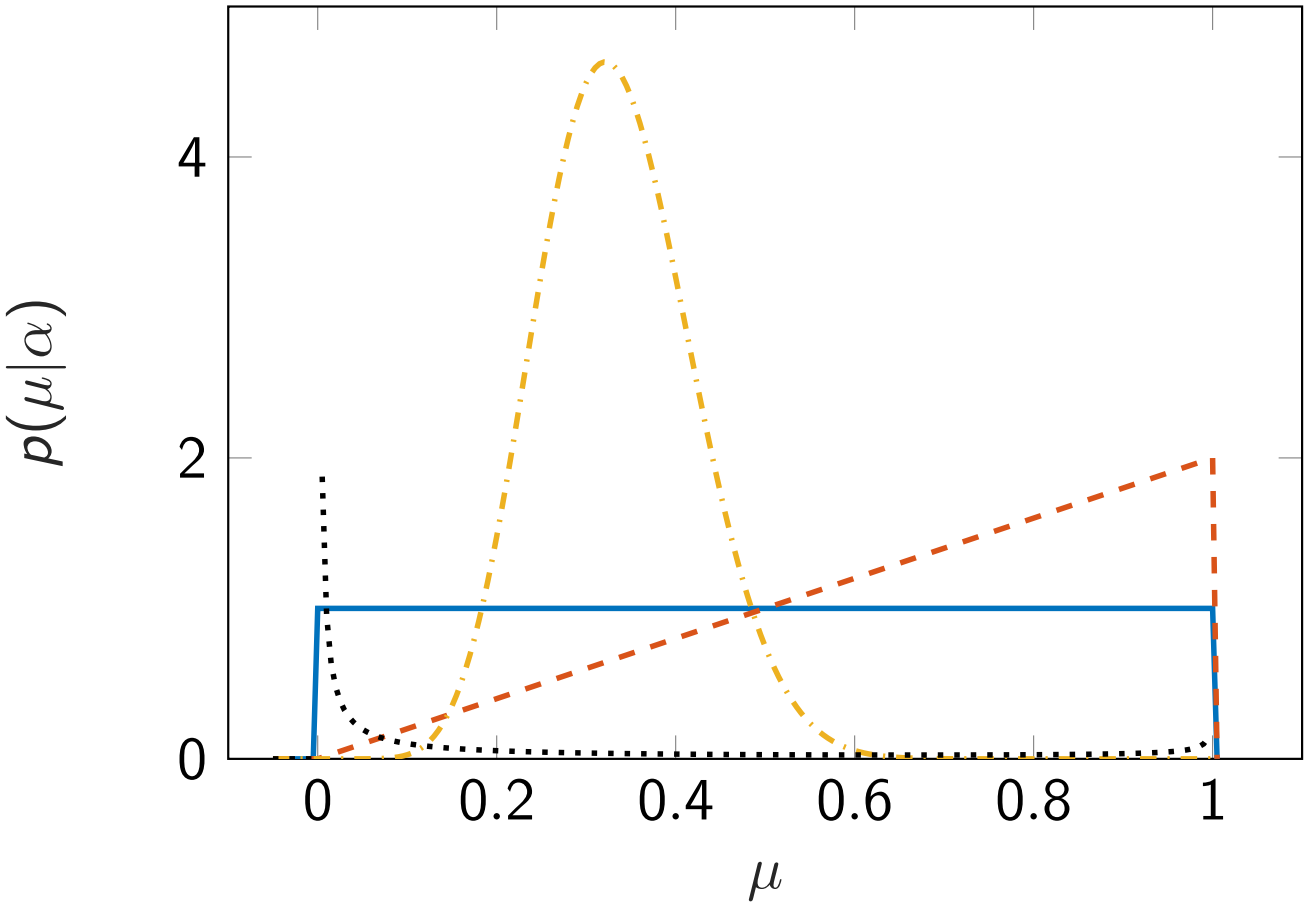


Figure: Four example Beta densities

# Beta prior for Bernoulli distributions

Figure: Beta-Bernoulli graphical model.

- We assume $\mu$ is unknown, but fixed.
- We observe $x^t = (x_1, \dots, x_t)$.
- Our prior density is determined by $\alpha$: $\xi_0(\mu) \triangleq p(\mu \mid \alpha)$.
- Our posterior probability is

$$\xi_t(\mu) = \frac{\prod_{k=1}^t P_\mu(x_k) p(\mu \mid \alpha)}{\int_0^1 \prod_{k=1}^t P_\mu(x_k) p(\mu \mid \alpha) \, \mathrm{d}\mu}$$

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$$\propto \mu^{s_{t,1}+\alpha_1-1}(1-\mu)^{s_{t,0}+\alpha_0-1}$$

where $s_{t,i} = \sum_{k=1}^t \mathbb{I}\{x_k = i\}$.

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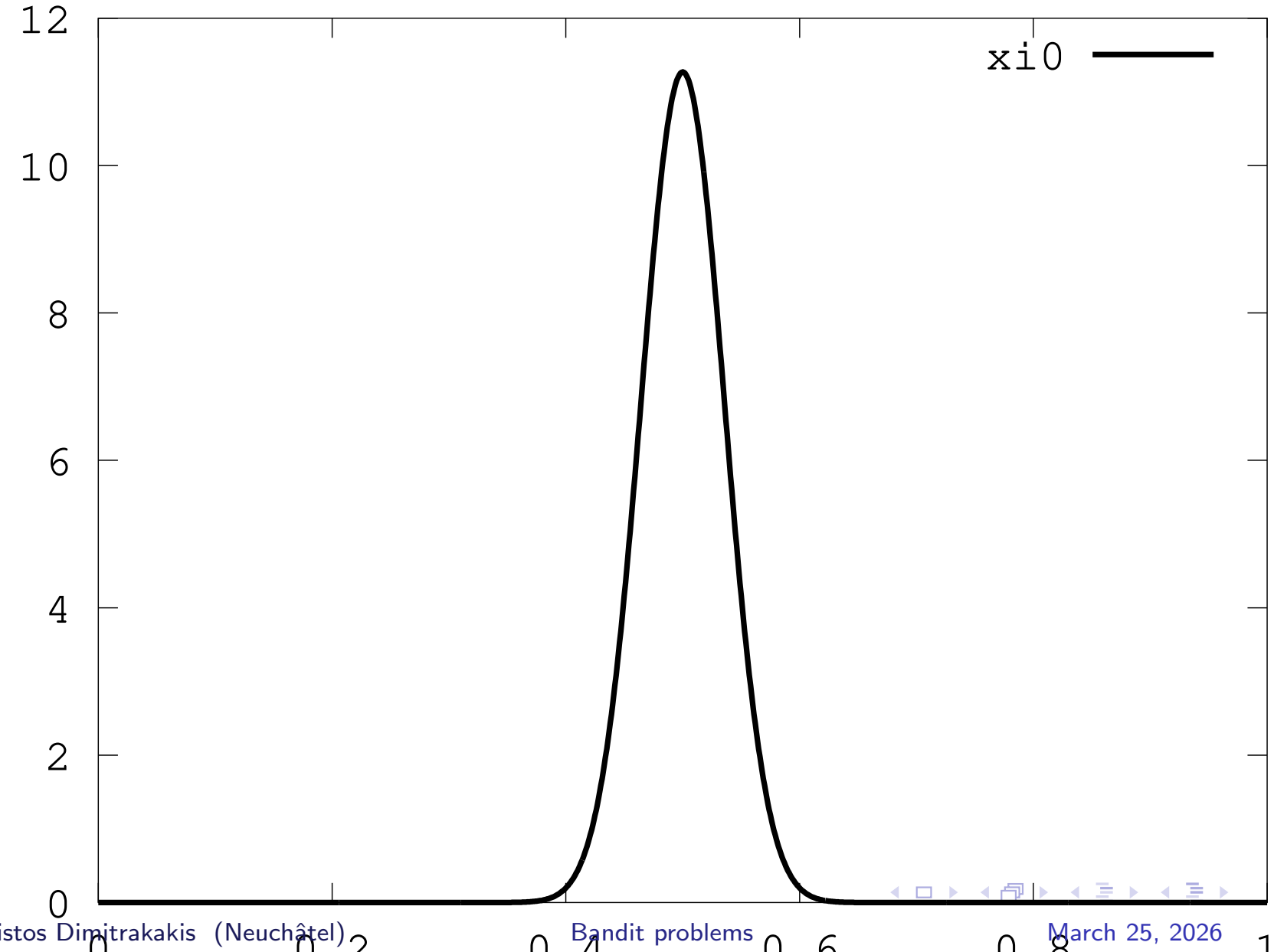
$$\begin{aligned}\xi_t(\mu) &= \frac{\prod_{k=1}^t P_\mu(x_k) p(\mu \mid \alpha)}{\int_0^1 \prod_{k=1}^t P_\mu(x_k) p(\mu \mid \alpha)\, \mathrm{d}\mu} \\ &\propto \mu^{s_{t,1}+\alpha_1-1}(1-\mu)^{s_{t,0}+\alpha_0-1} \\ &= f(\mu \mid \alpha + s).\end{aligned}$$

where $s_{t,i} = \sum_{k=1}^t \mathbb{I}\{x_k = i\}$.

# Coin tossing



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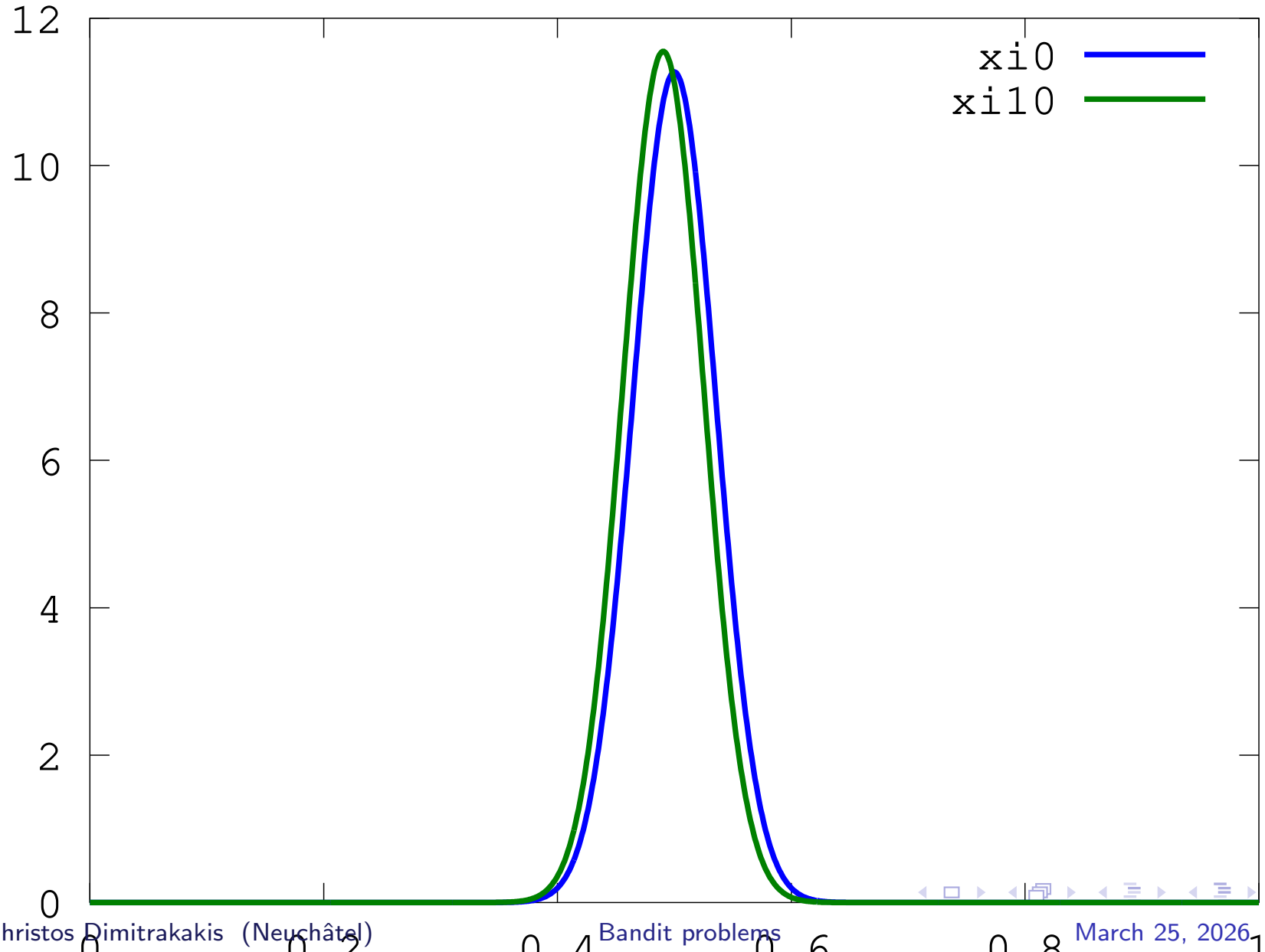

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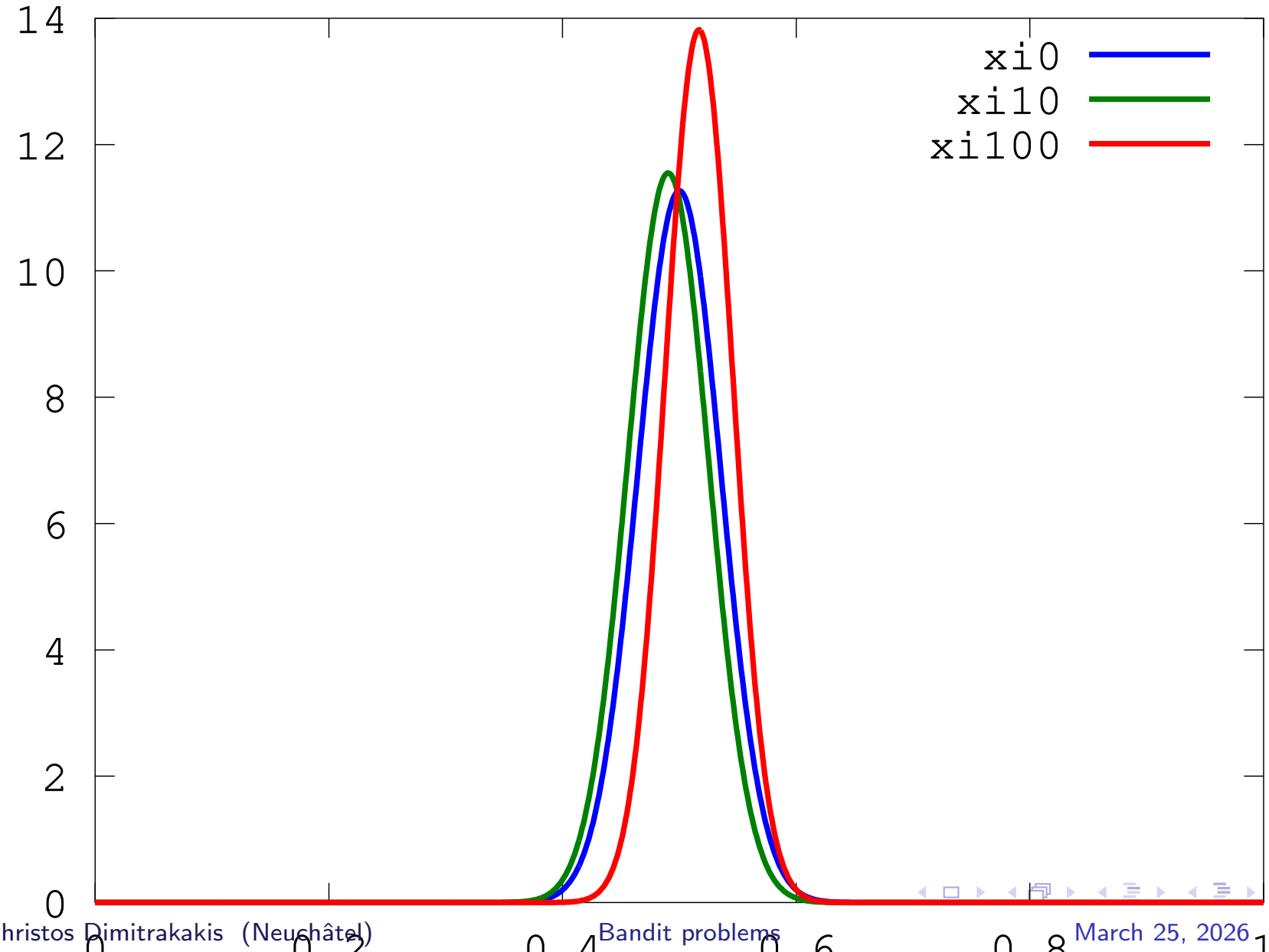

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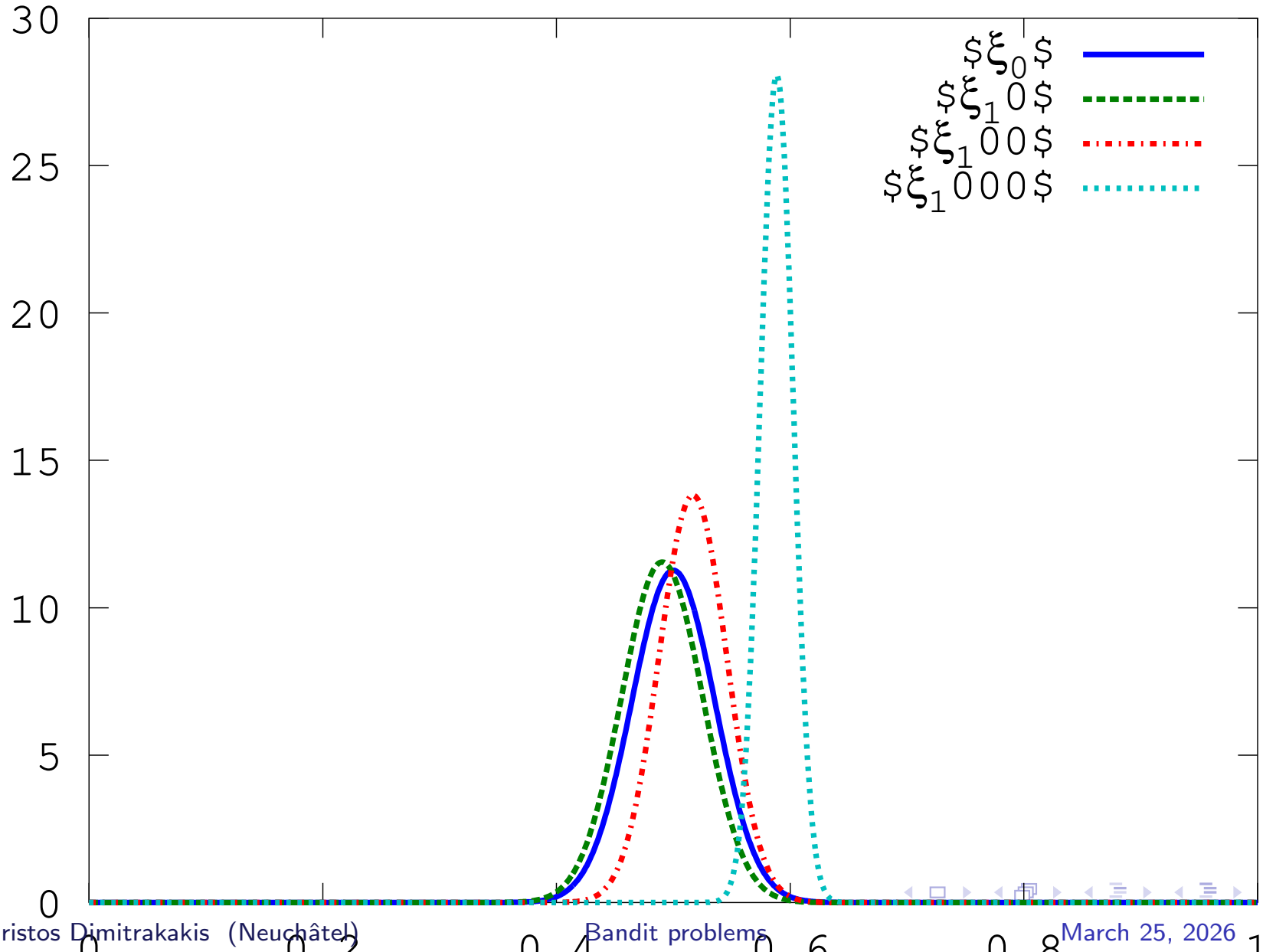
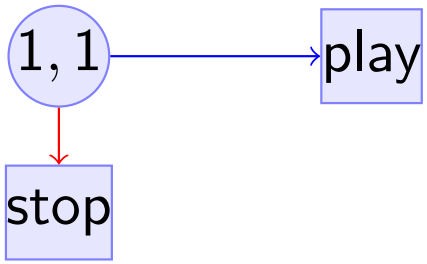


# one-armed Beta-Bernoulli bandit




$\xi_1$ $a_1$

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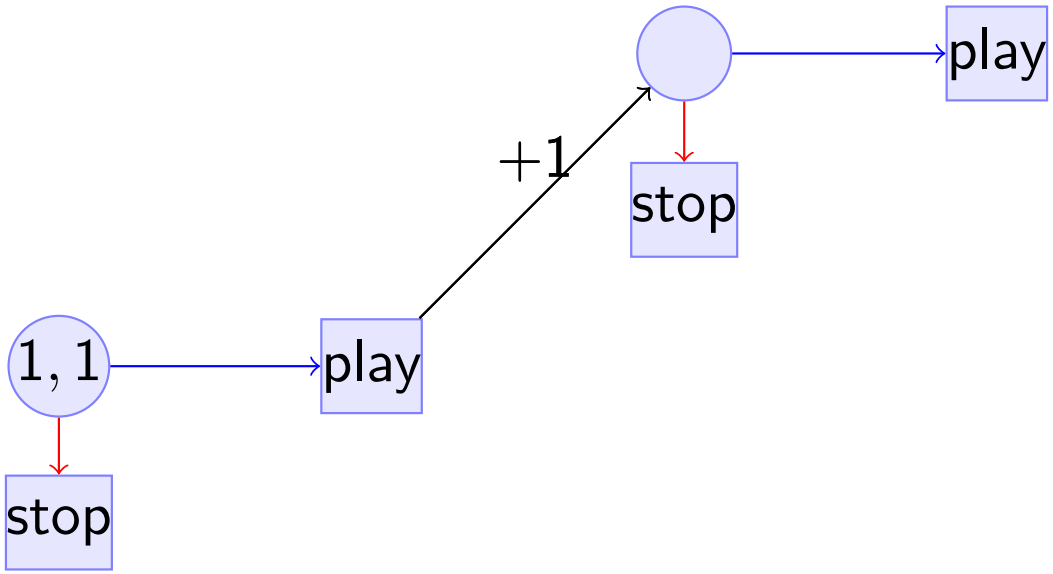


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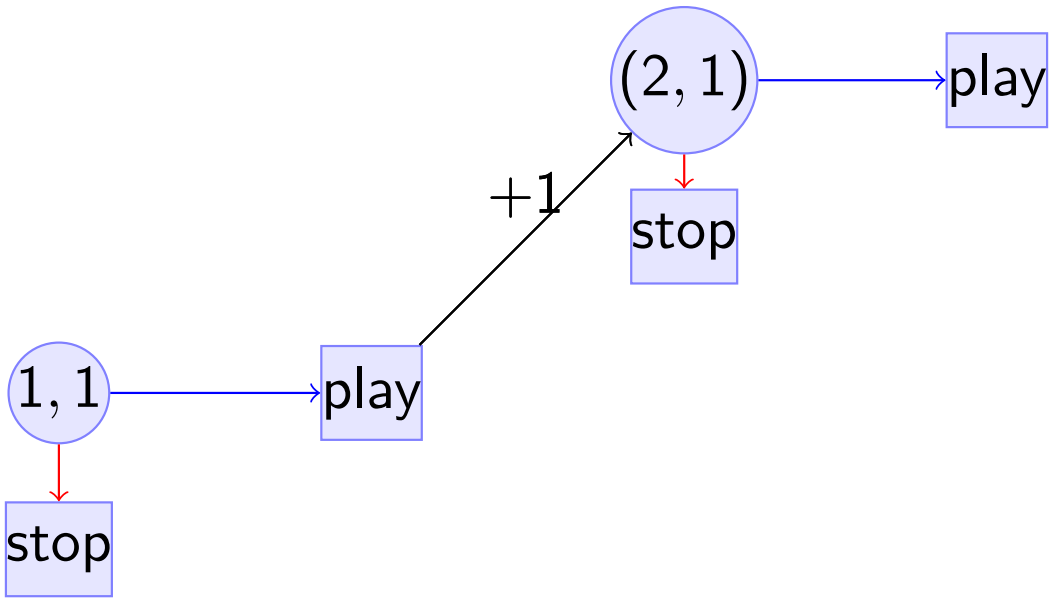


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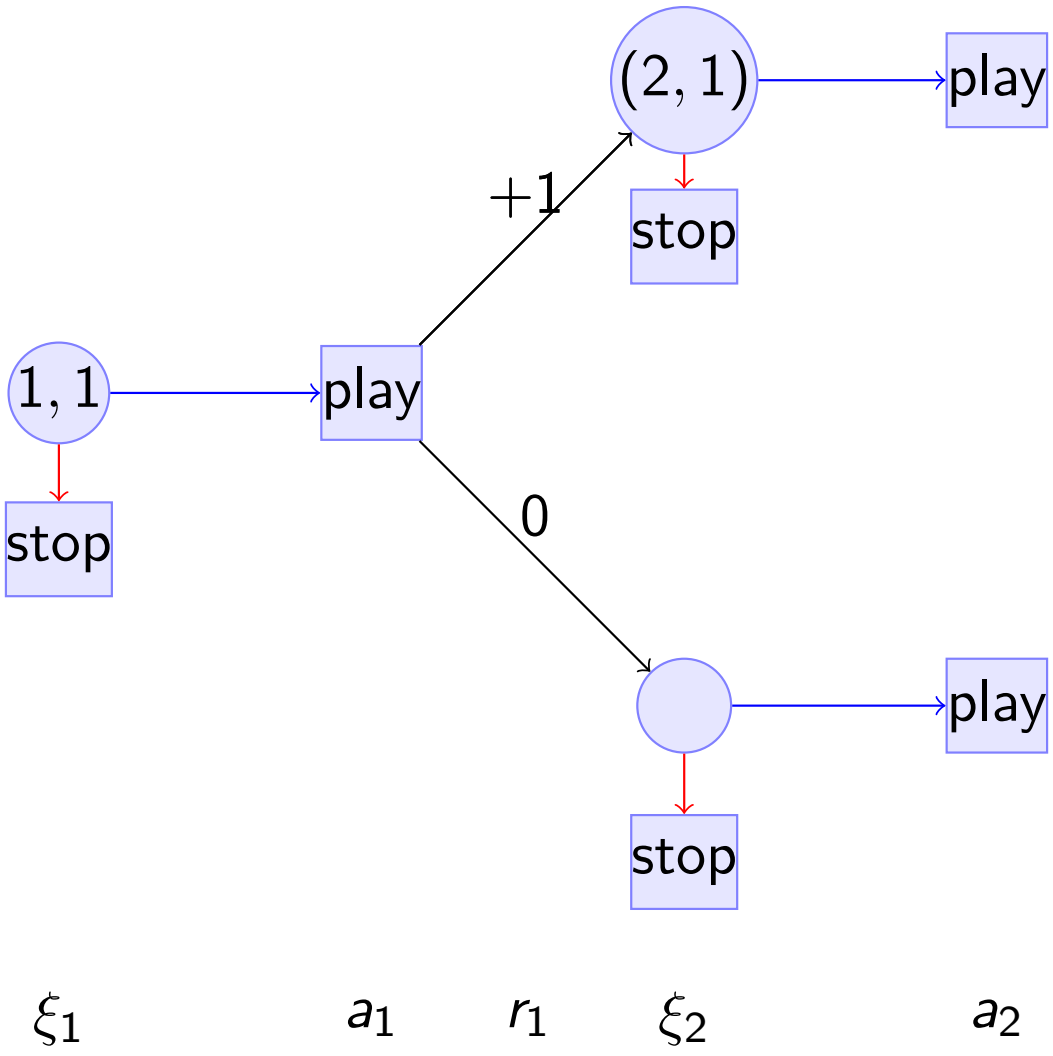

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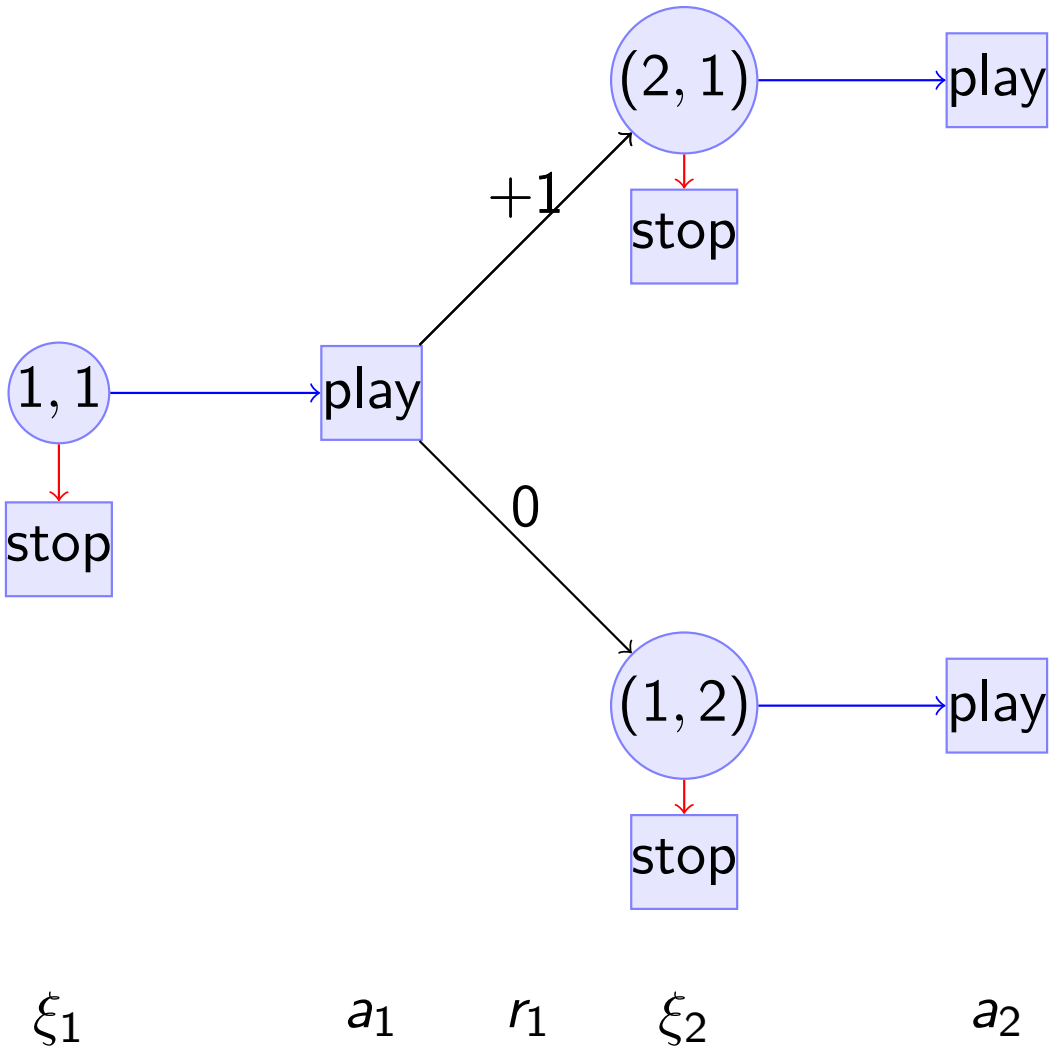

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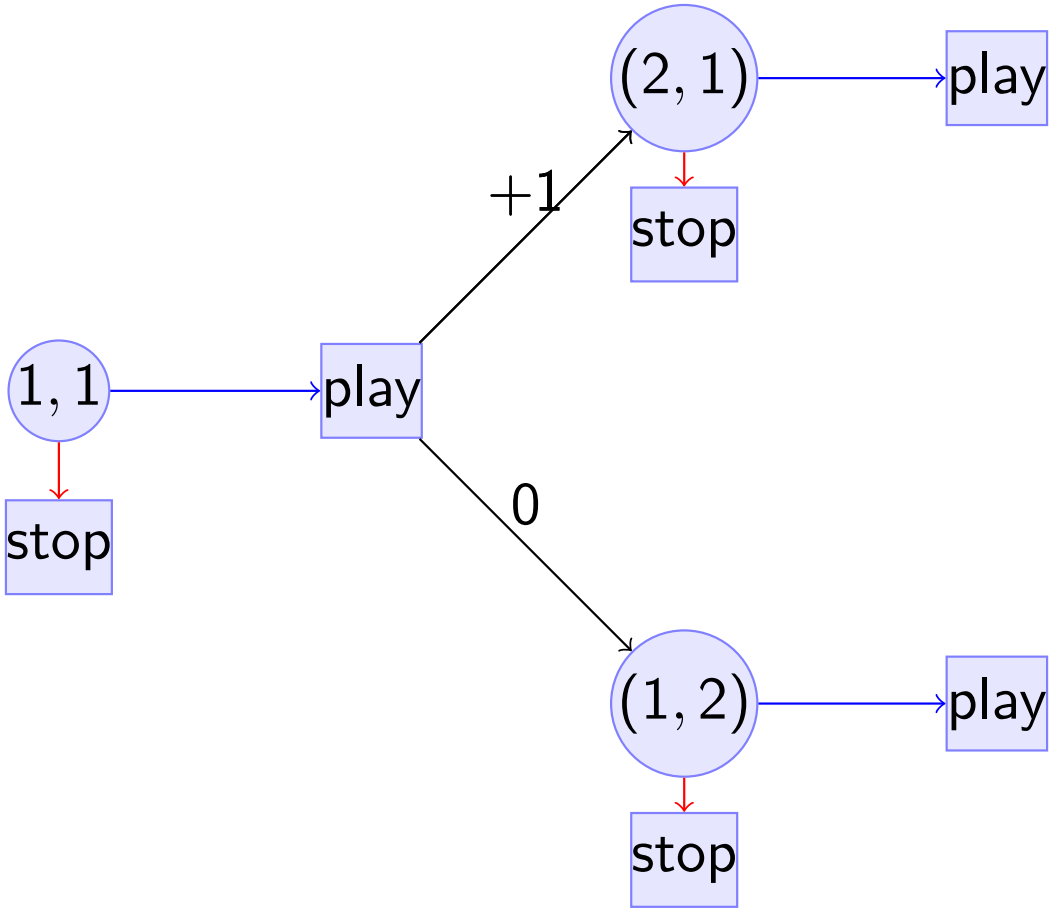
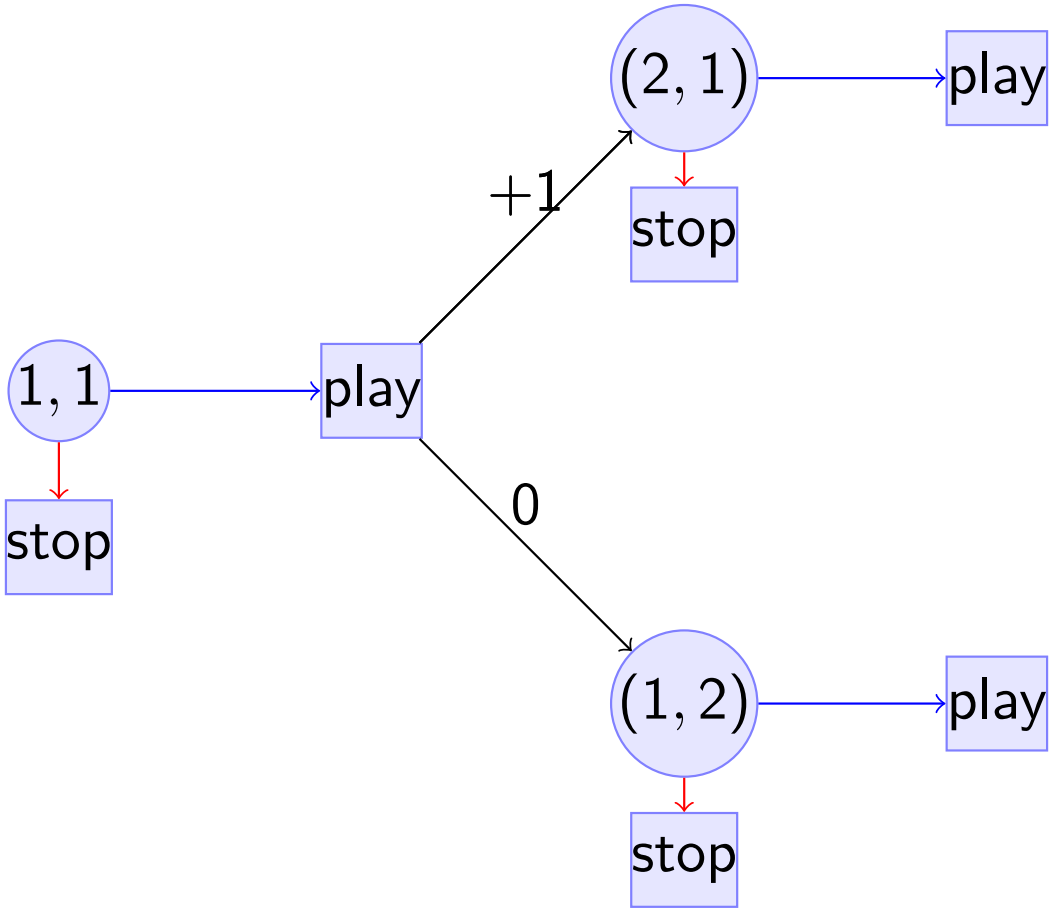



$\xi_1$ $a_1$ $r_1$ $\xi_2$ $a_2$

How can we generalise this to the $k$-armed problem?

# one-armed Beta-Bernoulli bandit




$\xi_1$ $a_1$ $r_1$ $\xi_2$ $a_2$

How can we approximate the optimal solution?

# The running average algorithm

Say you keep a running average of the reward obtained by each arm

$$\hat{\mu}_{t,i} = R_{t,i}/n_{t,i}$$

where $n_{t,i}$ is the number of times you played arm $i$ and $R_{t,i}$ the total reward received from $i$ so that whenever you play $a_t = i$:

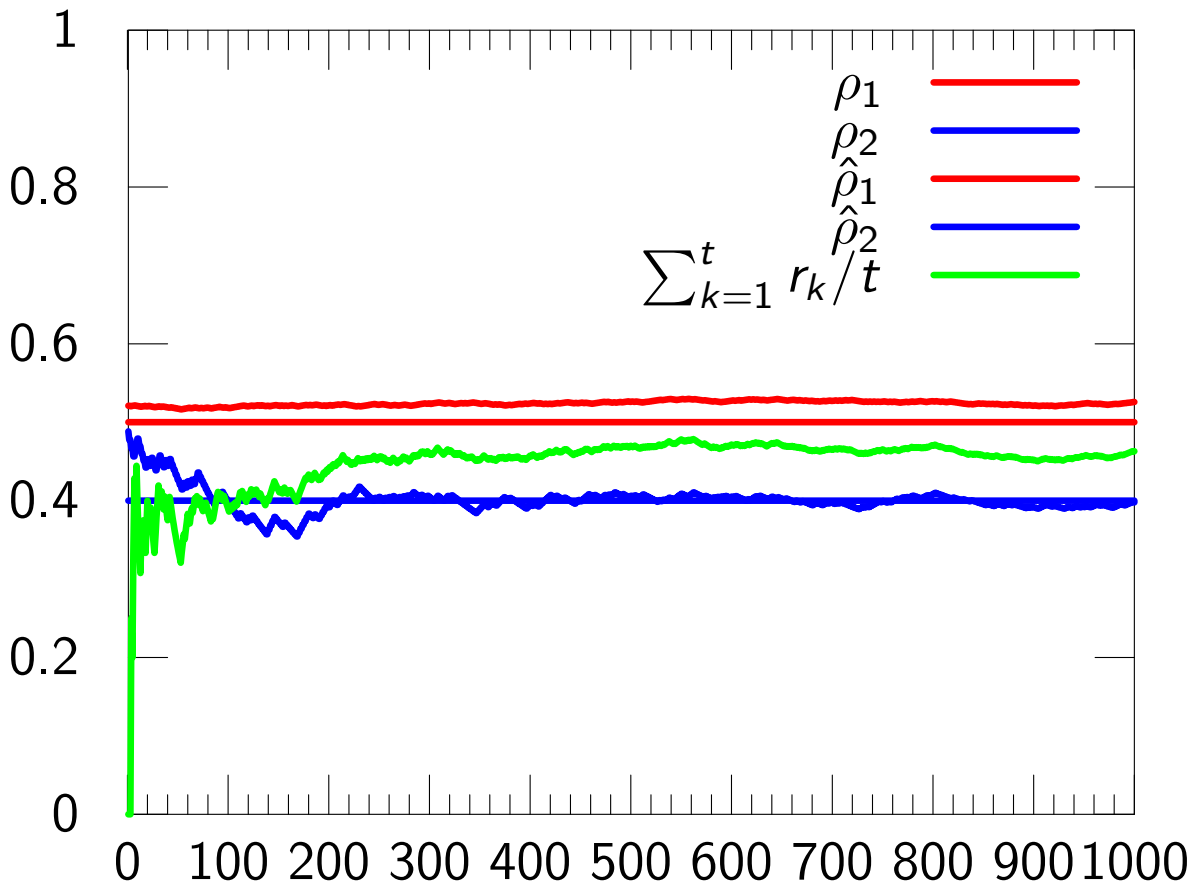
$$R_{t+1,i} = R_{t,i} + r_t, \qquad n_{t+1,i} = n_{t,i} + 1.$$

You could then choose to play the strategy

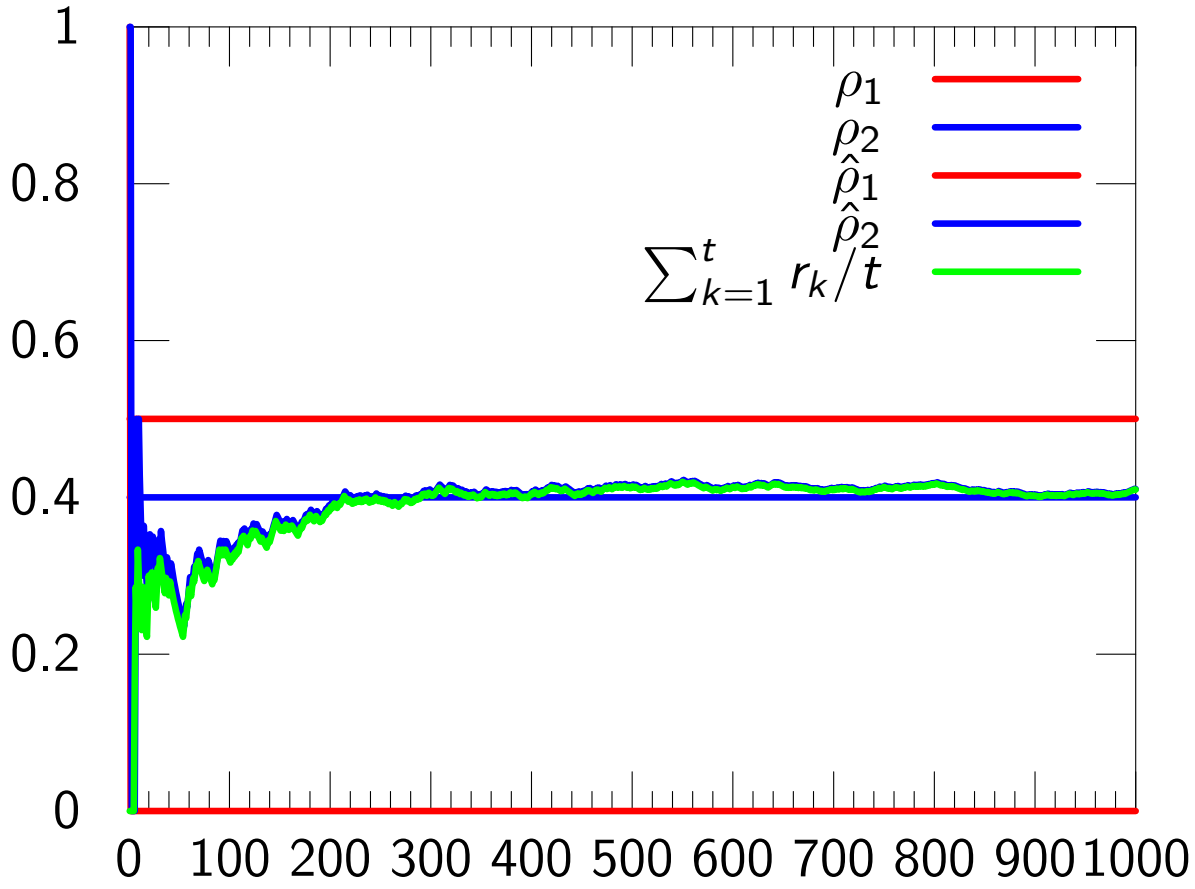
$$a_t = \arg\max_i \hat{\mu}_{t,i}.$$

What should the initial values $n_{0,i}, R_{0,i}$ be?

# The uniform policy

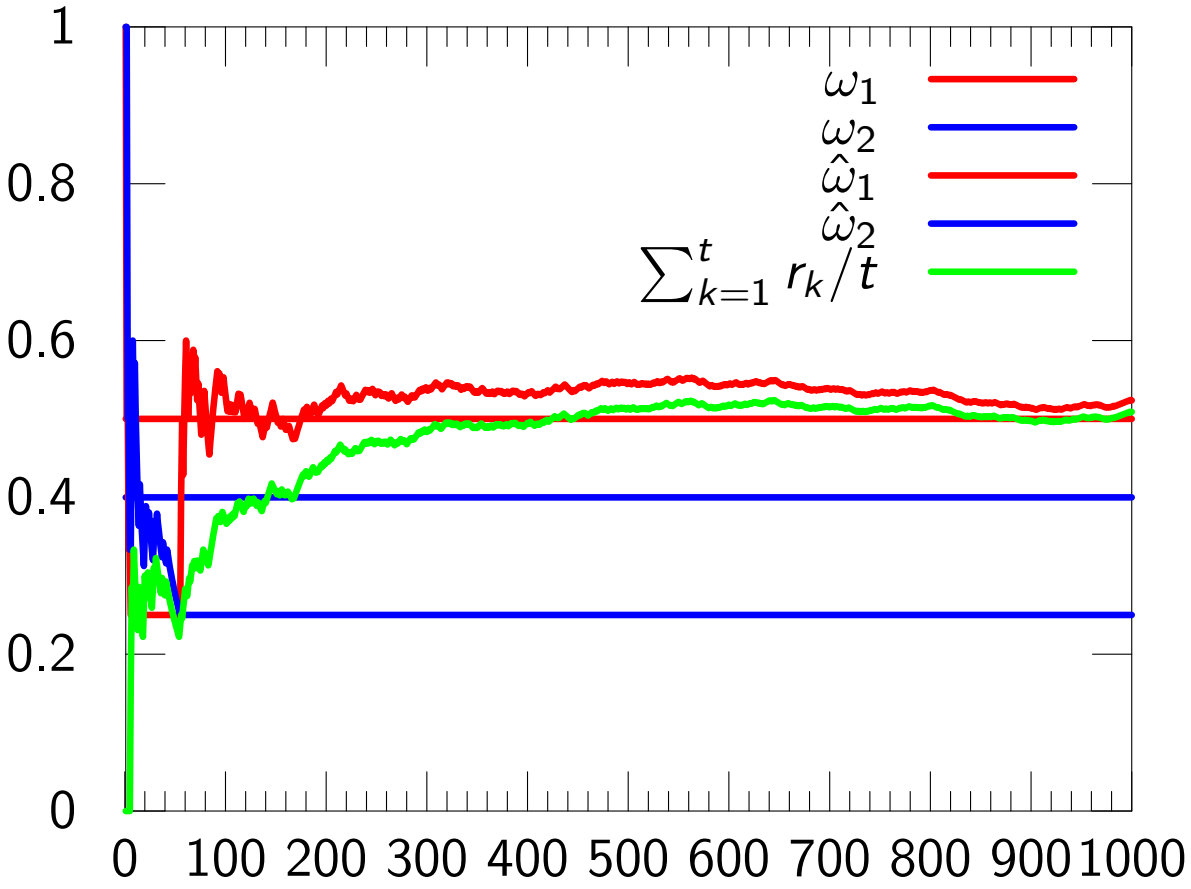


# The greedy policy



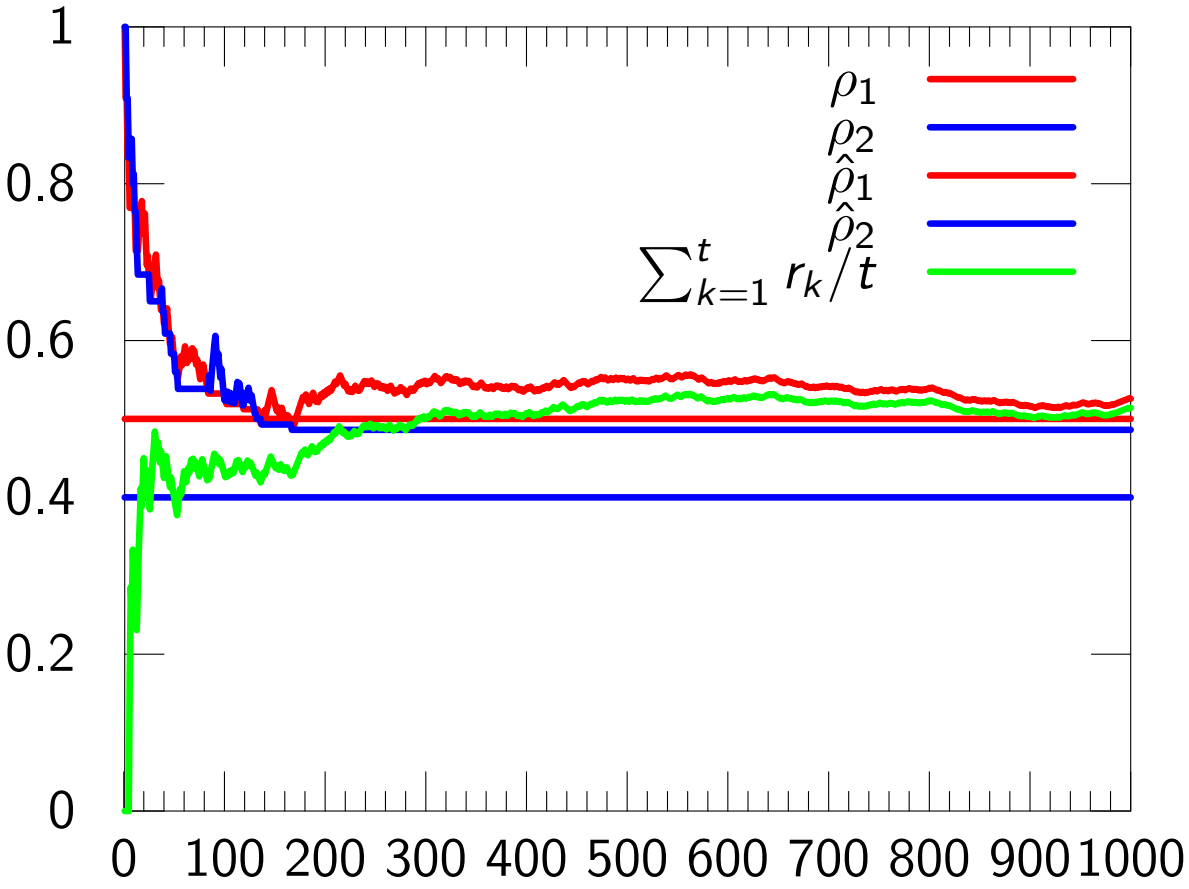

For $n_{0,i} = R_{0,i} = 0$

# The greedy policy




For $n_{0,i} = R_{0,i} = 1$

# The greedy policy




For $n_{0,i} = R_{0,i} = 10$

# Near-optimal heuristic policies

## Upper Confidence Bounds

Play the best average action plus an "exploration bonus"

$$a_t = \arg\max_i \hat{\mu}_{t,i} + \sqrt{\frac{2 \ln t}{n_{t,i}}}$$

## Posterior (Thompson) sampling

Sample the average values from the posterior and play the highest:

$$a_t = \arg\max_i \tilde{\mu}_{t,i}, \qquad \tilde{\mu}_{t,i} \sim \xi_t(\mu_i)$$

Both strategies are $O(1/T)$-optimal after $T$ plays.

# Optimal optimization

Consider the problem of finding the maximum of $f : \mathcal{X} \to \mathbb{R}$

$$x^* : f(x^*) \geq f(x) \forall x$$

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- We query points $x_t$ and observe a random $y_t$ s.t.

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What does this remind you of?

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How can we deal with the case when $\mathcal{X}$ is an infinite metric space?

- Discretise.
- Use a smooth approximation.
- Rely on Bayesian models to do all the smoothing.

# Active learning and experiment design

Consider the problem of finding a decision rule $f : \mathcal{X} \to \mathcal{Y}$ that (say) classifies instances into classes in $\mathcal{Y}$. What is the optimal way to do it?

- Say you can pay for $T$ labels.

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- So, build a tree to depth $T$...

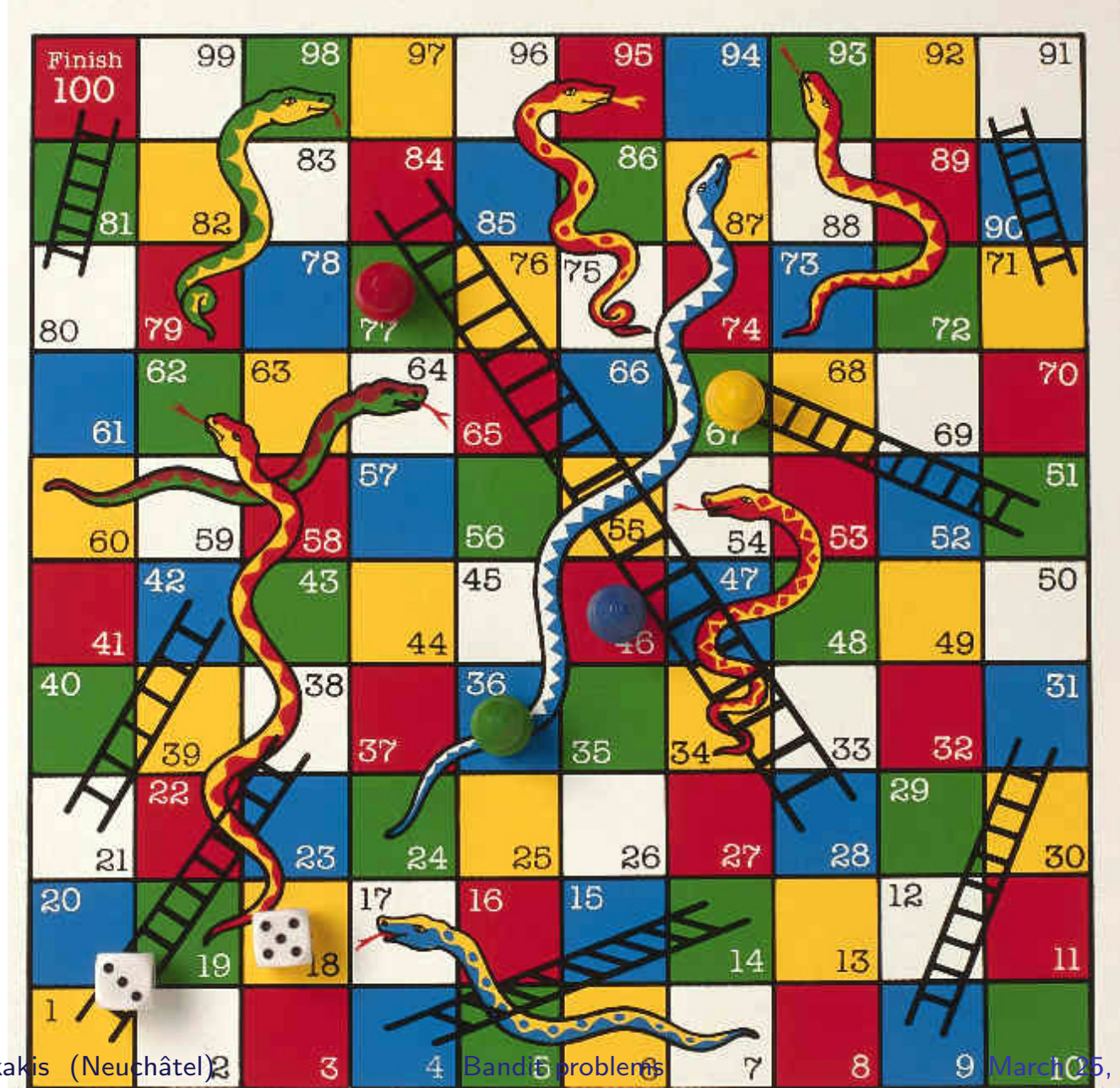
What's the size of this tree, approximately?

# Contents

Bandit problems
- Introduction
- Bernoulli bandits
- The unknown reward case
- Decision-theoretic bandit process
- Bernoulli-Beta conjugate pair

Markov decision processes and reinforcement learning
- Markov processes
- Markov decision processes

# A Markov process



# Markov process

$$s_{t-1} \longrightarrow s_t \longrightarrow s_{t+1}$$

## Definition 6 (Markov Process – or Markov Chain)

The sequence $\{s_t \mid t = 1, \ldots\}$ of random variables $s_t : \Omega \to \mathcal{S}$ is a Markov process if

$$\mathbb{P}(s_{t+1} \mid s_t, \ldots, s_1) = \mathbb{P}(s_{t+1} \mid s_t). \tag{2.1}$$

- $s_t$ is state of the Markov process at time $t$.
- $\mathbb{P}(s_{t+1} \mid s_t)$ is the transition kernel of the process.

## The meaning of "state"

- The state can refer to the state of our belief...
- .. or the state of a real dynamic environment.

# Reinforcement learning

Learning to act unknown environment, by interaction and reinforcement.

- The environment has a changing state $s_t$.
- The agents observes the state $s_t$ (simplest case).
- The agent takes action $a_t$.
- It receives rewards $r_t$.

## The goal (informally)

Maximise total reward $\sum_t r_t$

## Types of environments

- Markov decision processes (MDPs).
- Partially observable MDPs (POMDPs).
- (Partially observable) Markov games.

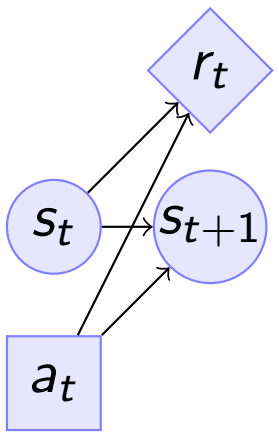
First deal with the case when $\mu$ is known.

# Markov decision processes

## MDP

At each time step $t$:

- We observe state $s_t \in \mathcal{S}$.
- We take action $a_t \in \mathcal{A}$.
- We receive a reward $r_t \in \mathbb{R}$.




## Markov property of the reward and state distribution

$$\mathbb{P}_\mu(s_{t+1} \mid s_t, a_t) \qquad \text{(Transition distribution)}$$

$$\mathbb{P}_\mu(r_t \mid s_t, a_t) \qquad \text{(Reward distribution)}$$

# The agent

## The agent's policy $\pi$

$$\mathbb{P}^\pi(a_t \mid s_t, \dots, s_1, a_{t-1}, \dots, a_1) \qquad \text{(history-dependent policy)}$$

$$\mathbb{P}^\pi(a_t \mid s_t) \qquad \text{(Markov policy)}$$

Given a horizon $T$, the utility can be defined as

$$U_t \triangleq \sum_{k=0}^{T-t} r_{t+k} \tag{2.2}$$

The agent wants to to find $\pi$ maximising the expected total future reward

$$\mathbb{E}_\mu^\pi U_t = \mathbb{E}_\mu^\pi \sum_{k=0}^{T-t} r_{t+k}. \qquad \text{(expected utility)}$$

# A teaser

Backwards-induction algorithms apply to any MDP. For special cases, two other exact algorithms exist.
But more on those next time...

# Conclusion

- Externally, a bandit process is an MDP with just one fixed state.
- From the viewpoint of a learning agent, it is an MDP as well, whose state is the agent's belief.
- A dynamic environment has a dynamic (observable?) state, affected by the agent. It is also an MDP.
- As we shall see, a learning agent can combine its belief and the environment state into a hyperstate to form a (bigger) MDP.

These MDPs are pretty big.

- How much effort do we need to solve them?
- How much data do we need to solve them?

# Exercises

1. Compare a number of heuristic strategies on a simple bandit problem (at least UCB, Thompson sampling and the average). Feel free to consult external sources! Use a Bernoulli bandit with $k = 8$ arms, with the $i$-th arm obtaining average reward $i/(k+1)$.
2. Try and precisely formulate the optimal optimization problem as a bandit problem. How can you interpret it?
3. What should the utility function for active learning be?
4. Is there some essential difference between MDPs describing a bandit process and MDPs describing a dynamic environment?

Use the ILIAS website to discuss algorithms and co-ordinate.

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